THREAT HUNT REPORT

APT41 (Double Dragon) — TTP Simulation & Detection

Using Splunk SIEM on Windows 10 Lab Environment

Author: Ishaan Malhotra
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1. Executive Summary

This report documents a threat hunt exercise conducted against simulated APT41 (Double Dragon) attack techniques in a controlled Windows 10 lab environment. APT41 is a Chinese state-sponsored threat actor known for conducting both espionage operations and financially motivated cybercrime.

Six APT41 TTPs were simulated using the Atomic Red Team framework. Three were successfully detected using Splunk SIEM with default Windows Event Log configuration. Three techniques went undetected due to insufficient logging — a key finding that highlights critical gaps in default Windows logging.

Metric	Result
TTPs Simulated	6
TTPs Detected	3
TTPs Missed	3
Detection Rate	50%
SIEM Used	Splunk Enterprise 10.2.1
Lab Environment	Windows 10 VM (VMware)

2. Threat Actor Profile — APT41

Attribute	Details
Group Name	APT41 (Double Dragon)
Also Known As	Winnti, Barium, LEAD
Origin	China (PLA-linked)
Active Since	2012
Motivation	Espionage + Financial Gain
Primary Targets	Healthcare, Telecom, Gaming, Finance, Government
Known Malware	DUSTPAN, DEADEYE, Winnti, HIGHNOON
MITRE ID	G0096

APT41 is unique among Chinese threat actors in that it conducts both state-sponsored espionage and independent financially motivated operations — including ransomware deployment and supply chain attacks. This dual mandate makes them one of the most versatile and dangerous APT groups tracked globally.

3. Hunt Hypothesis

Based on APT41's known TTPs documented in MITRE ATT&CK (G0096), the following hypothesis was formed:

> *"APT41 may have established persistence on Windows endpoints through malicious service installation (T1543.003) and created backdoor accounts (T1136) while abusing elevated privileges (T1078) to maintain long-term access."*

Six TTPs were selected for simulation based on APT41's documented behavior in the MITRE ATT&CK framework:

#	MITRE ID	Technique	Why Selected
1	T1059.001	PowerShell Execution	APT41's primary execution method
2	T1053.005	Scheduled Task	Common persistence mechanism
3	T1543.003	Windows Service	APT41 uses fake services for DUSTPAN
4	T1078	Valid Account Abuse	Privilege escalation via stolen credentials
5	T1560.001	RAR Data Archiving	APT41 compresses data before exfiltration
6	T1136	Account Creation	Backdoor account for persistent access

4. Lab Environment

Component	Details
Host OS	Windows 10 x64
Virtualization	VMware Workstation
SIEM	Splunk Enterprise 10.2.1 (Free Trial)
Attack Simulation	Atomic Red Team (Invoke-AtomicRedTeam)
Log Sources	Windows Security, System, Application Event Logs
Network	Isolated VM — no internet exposure

4.1 Lab Setup Steps

- Deployed Windows 10 VM in VMware
- Installed Splunk Enterprise and configured local Windows Event Log ingestion
- Disabled Windows Defender real-time monitoring for simulation purposes
- Installed Invoke-AtomicRedTeam framework and downloaded all atomics
- Executed 6 APT41 TTP simulations using Atomic Red Team
- Investigated generated logs in Splunk using targeted SPL queries

5. Evidence & Findings

5.1 DETECTED — T1543.003: Windows Service Creation

Field	Value
EventCode	7045
Log Source	WinEventLog:System
Findings	2 malicious services detected
Service Names	AtomicTestService_PowerShell, AtomicTestService_CMD
File Path	C:\AtomicRedTeam\atomic-red-team-master\atomics\T1543.003\bin\AtomicService.exe
Start Type	Auto Start
Account	LocalSystem
Severity	HIGH

APT41 installed two fake Windows services set to auto-start under the LocalSystem account. This is consistent with APT41's documented use of Windows services for DUSTPAN malware persistence. The suspicious service file path outside of System32 is a strong indicator of compromise.

```
SPL Query: index=* EventCode=7045 | table _time ComputerName Message
```

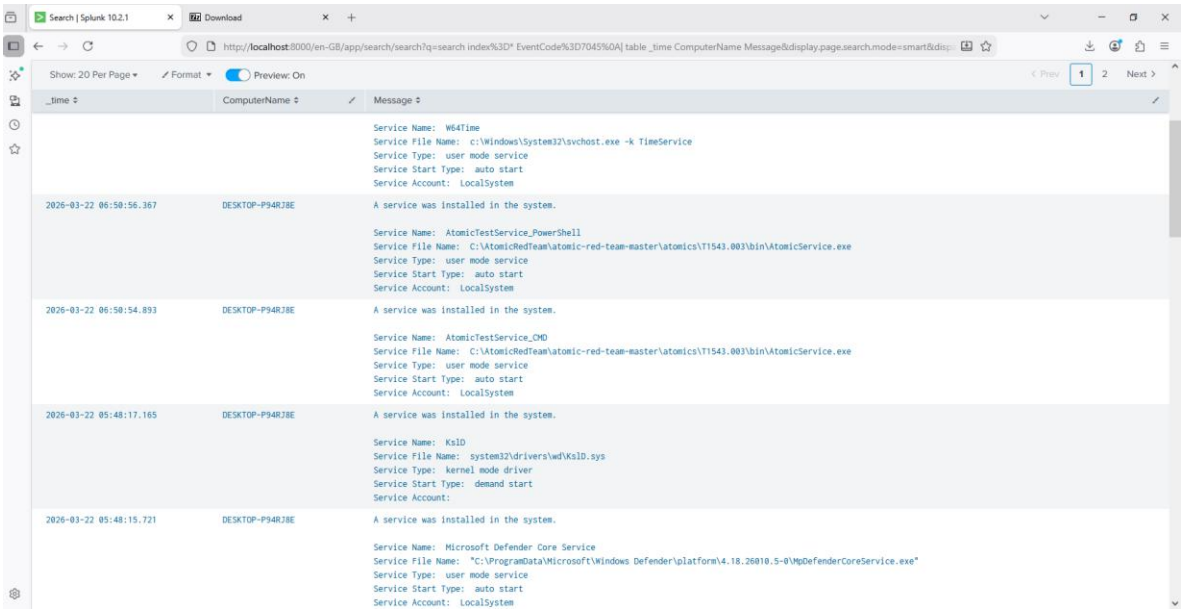

Figure 1: Splunk detection of malicious service installation (EventCode 7045)

5.2 DETECTED — T1078: Privilege Escalation via Special Logon

Field	Value
EventCode	4672
Log Source	WinEventLog:Security
Findings	348 events detected
Account	SYSTEM (NT AUTHORITY)
Key Privileges	SeDebugPrivilege, SeImpersonatePrivilege, SeTakeOwnershipPrivilege
Severity	HIGH

348 instances of special privilege assignment were detected during the simulation window. The presence of SeDebugPrivilege and SeImpersonatePrivilege are well-known privilege abuse indicators associated with credential theft and lateral movement — core APT41 behaviors.

```
SPL Query: index=* EventCode=4672 | table _time Account_Name Message
```

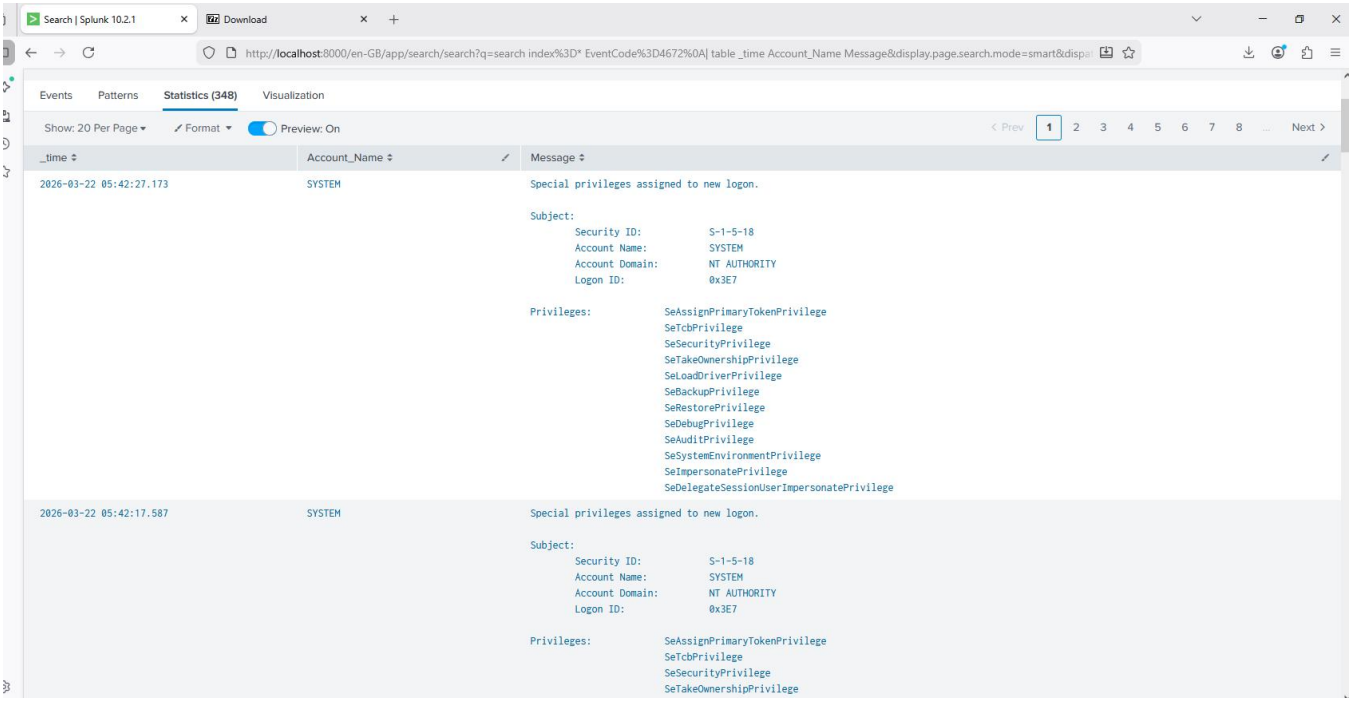

Figure 2: Splunk detection of privilege abuse — 348 events (EventCode 4672)

5.3 DETECTED — T1136: Backdoor Account Creation

Field	Value
EventCode	4720
Log Source	WinEventLog:Security
Findings	3 account creation events
Account Created	kali
Created By	SYSTEM (S-1-5-18)
Password Required	No — suspicious
Severity	CRITICAL

A backdoor account named 'kali' was created by the SYSTEM account with no password required. Account creation by SYSTEM rather than a named administrator is a strong indicator of automated malicious activity. This is consistent with APT41's documented practice of creating persistence accounts for long-term access.

```
SPL Query: index=* EventCode=4720 | table _time ComputerName TargetUserName Message
```

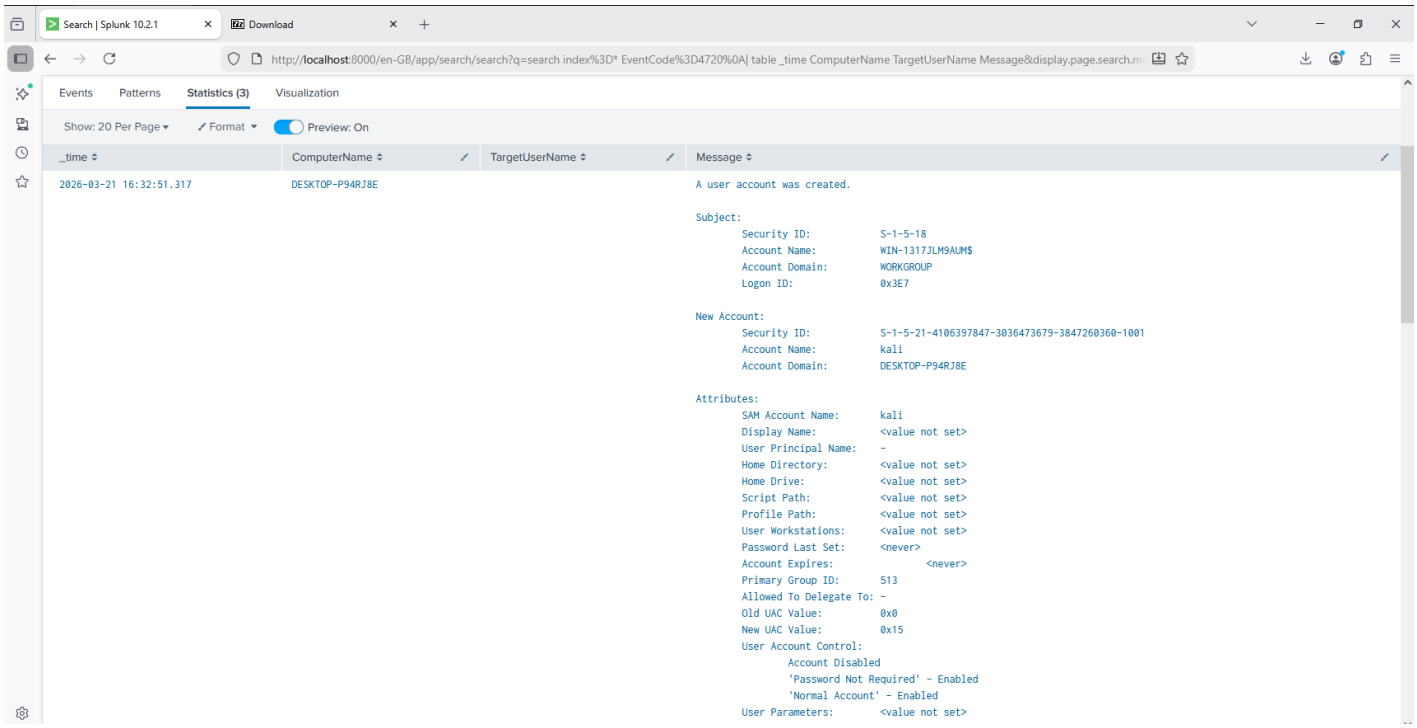

Figure 3: Splunk detection of backdoor account creation (EventCode 4720)

6. Missed Detections & Root Cause

Three TTPs went undetected due to Windows logging gaps. The attacks DID occur — Splunk simply had no log data to analyze because Windows did not record the activity by default.

TTP	Technique	Why Missed	Fix
T1059.001	PowerShell Execution	Script Block Logging disabled by default	Enable via Group Policy
T1053.005	Scheduled Task Creation	Task Scheduler log disabled by default	Enable via wevtutil
T1560.001	RAR Data Archiving	No file activity monitoring configured	Deploy Sysmon Rule 11

This finding highlights a critical gap: default Windows logging is insufficient for threat hunting. Organizations relying solely on default Windows Event Logs will miss a significant portion of APT41's attack chain.

7. MITRE ATT&CK Mapping

MITRE ID	Technique	Tactic	Detected	EventCode
T1059.001	PowerShell	Execution	No	4104 (disabled)
T1053.005	Scheduled Task	Persistence	No	4698 (disabled)
T1543.003	Windows Service	Persistence	Yes	7045
T1078	Valid Accounts	Privilege Escalation	Yes	4672
T1560.001	Data Archiving	Collection	No	N/A
T1136	Account Creation	Persistence	Yes	4720

8. Recommendations

8.1 Immediate Actions

- Enable PowerShell Script Block Logging via Group Policy to detect T1059.001
- Enable Task Scheduler Operational logs via wevtutil to detect T1053.005
- Deploy Sysmon with SwiftOnSecurity ruleset for comprehensive process and file monitoring
- Alert on any new service installation outside of System32 (EventCode 7045)
- Alert on new local account creation by SYSTEM account (EventCode 4720)

8.2 Long-Term Improvements

- Deploy an EDR solution (e.g., Microsoft Defender for Endpoint) for deeper telemetry
- Implement UEBA to detect anomalous privilege usage patterns
- Regularly update detection rules based on APT41's evolving TTPs from MITRE ATT&CK
- Conduct quarterly threat hunt exercises against updated APT41 intelligence
- Enable cloud storage monitoring to detect T1567.002 exfiltration to OneDrive

9. Conclusion

This threat hunt exercise successfully demonstrated the ability to detect key APT41 persistence and privilege escalation techniques using Splunk SIEM with Windows Event Logs. Three of six simulated TTPs were detected, yielding a 50% detection rate with default logging configuration.

The three missed detections are not failures — they are actionable findings. Each missed detection maps directly to a logging gap that can be remediated with specific, low-cost configuration changes. This is the true value of threat hunting: identifying blind spots before a real attacker exploits them.

The findings from this exercise demonstrate that default Windows logging is insufficient for detecting sophisticated APT41 intrusions. Organizations must invest in enhanced logging, Sysmon deployment, and EDR solutions to achieve adequate visibility against this threat actor.

Item	Detail
Report Author	Ishaan Malhotra
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Tools Used	Splunk, Atomic Red Team, VMware, Windows 10
Reference	MITRE ATT&CK G0096 — APT41